



HERS-VERIFIED SINGLE DWELLING UNIT HOT WATER SYSTEM DISTRIBUTION

CALIFORNIA ENERGY COMMISSION

CEC-CF3R-PLB-22-H

SAMPLE FORM – NOT VALID FOR SUBMISSION TO BUILDING DEPARTMENTS

CERTIFICATE OF VERIFICATION

Note: This table completed by HERSECC Registry.

Project Name:	Enforcement Agency:
Dwelling Address:	Permit Number:
City and Zip Code:	Permit Application Date:

A. Design **HERS**Verified Dwelling Unit Water Heating Systems Information (other than HPWH)

This table reports features of the water heating system(s) other than HPWH system specified on the registered CF1R compliance document for this project.

01	02	03	04	05	06	07	08	10	11	12
Dwelling Unit Name	Water Heating System ID or Name	Water Heating System Type	Water Heater Type	# of Like (or Identical) Water Heaters in System	Fuel Type	Rated Input Type	Rated Input Value	Dwelling Unit DHW System Distribution Type	Compact Distrib.	Drain Water Heat Recovery

A2. Design **HERS**Verified Dwelling Unit HPWH System Information

This table reports the water heating system(s) that were specified on the registered CF1R compliance document for this project.

01	02	03	04	05	06	06a	07	08	09	10
Dwelling Unit Name	Water Heating System ID or Name	Modeled Equipment Make and Model	# of Like (or Identical) Water Heaters in System	Tank Location	Exterior Tank Insulation R-value	Tank Volume	Dwelling Unit DHW System Distribution Type	Compact Distribution	Drain Water Heat Recovery	Simulated Equipment Make and Model

B. Installed **HERS**Verified Dwelling Unit Water Heating Systems Information

This table reports the water heating system features installed in this project.

01	02	03	04	05	06	07	08	10	11	12
Dwelling Unit Name	Water Heating System ID or Name	Water Heating System Type	Water Heater Type	# of Like (or Identical) Water Heaters in System	Fuel Type	Rated Input Type	Rated Input Value	Dwelling Unit DHW System Distribution Type	Compact Distrib.	Drain Water Heat Recovery

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B2. Installed **HERS**Verified Dwelling Unit HPWH System Information

This table reports the water heating system(s) installed in this project.

01	02	03	04	05	06	06a	07	08	09
Dwelling Unit Name	Water Heating System ID or Name	Modeled Equipment Make and Model	# of Like (or Identical) Water Heaters in System	Tank Location	Exterior Tank Insulation R-value	Tank Volume	Dwelling Unit DHW System Distribution Type	Compact Distribution	Drain Water Heat Recovery

C. Design **HERS**Verified Dwelling Unit Water Heating Efficiency Information

This table reports the water heater(s) efficiency features specified on the registered CF1R compliance document for this project. (Not needed for central systems)

01	02	03	04	05	06	07
Water Heating System ID or Name	Heating Efficiency Type	Heating Efficiency Value	Standby Loss (%)	Exterior Insulation R-Value	Water Heater Storage Volume (gal)	Tank location

D. Installed **HERS**Verified Dwelling Unit Water Heating Efficiency Information

This table reports the water heater(s) efficiency features installed in this project. (Not needed for central systems)

01	02	03	04	05	06	07
Water Heating System ID or Name	Heating Efficiency Type	Heating Efficiency Value	Standby Loss (%)	Exterior Insulation R-Value	Water Heater Storage Volume (gal)	Tank location
08	Compliance Statement					

E. Installed Water Heater Manufacturer Information

01	02	03
Water Heating System ID or Name	Manufacturer	Model Number

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F. Mandatory Measures for ~~all Domestic Hot Water Distribution~~ Single Dwelling Systems

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met unless otherwise noted in the Verification Status and the Corrections Notes in this table.

01	Equipment shall meet the applicable requirements of the Appliance Efficiency Regulations (Section 110.3(b)1).
02	Unfired Storage Tanks are insulated with an external R-3.5 or combination of R-16 internal and external Insulation. (Section 110.3(c) 3 4).
03	• All domestic hot water piping shall be insulated as specified in Section 609.12 of the California Plumbing Code. Insulation buried below grade must be installed in a waterproof and non-crushable casing or sleeve. • Pipe insulation shall fit tightly and all elbows and tees shall be fully insulated. • Piping that penetrates framing members shall not be required to have pipe insulation for the distance of the framing penetration. • Piping that penetrates metal framing shall use grommets, plugs, wrapping or other insulating material to assure that no contact is made with the metal framing. Insulation shall butt securely against all framing members. • Piping surrounded with a minimum of 1 inch of wall insulation, 2 inches of crawlspace insulation, or 4 inches of attic insulation shall not be required to have pipe insulation. The following pipes are insulated, to the thicknesses required by Table 120.3A, except for those sections of pipe that are subject to one of the exceptions below: All domestic hot water piping shall be insulated as specified in Section 609.11 of the California Plumbing Code. Insulation buried below grade must be installed in a water-proof and non-crushable casing or sleeve. • Piping that penetrates framing members shall not be required to have pipe insulation for the distance of the framing penetration. Piping that penetrates metal framing shall use grommets, plugs, wrapping or other insulating material to assure that no contact is made with the metal framing. Insulation shall butt securely against all framing members. • Piping installed in interior or exterior walls that is surrounded on all sides by at least 1 inch (2.5 cm) of insulation. • Piping installed in crawlspace with a minimum of 1 inches (2.5 cm) of crawlspace insulation above and below. • Piping installed in attics with a minimum of 4 inches (10 cm) of attic insulation on top • Pipe insulation shall fit tightly and all elbows and tees shall be fully insulated.
04	For Gas or Propane Water Heaters: Ensure either a or b are installed (Section 150.0(n)) a) A designated space at least 2.5 feet by 2.5 feet and 7 feet tall within 3 feet from the water heater • A dedicated 125V, 20A electrical receptacle connected to the electric panel with a 120/240V 3 conductor, 10 AWG copper branch circuit <u>rated at 30 amps minimum</u>, within 3 feet from the water heater and is accessible with no obstructions; ▪ The conductor shall be labeled with the word "Spare" on both ends; and ▪ A reserved single pole circuit breaker space next to the circuit breaker next to the branch circuit labeled "Future" 240V <u>use</u> shall be provided. b) A designated space at least 2.5 feet by 2.5 feet and 7 feet tall more than 3 feet from the water heater • A dedicated 240 volt branch circuit shall be installed within 3 feet from the designated space. The branch circuit shall be rated at 30 amps minimum. The blank cover shall be identified as "240V ready"; and • The main electrical service panel shall have a reserved space to allow for the installation of a double pole circuit breaker for a future HPWH installation. The reserved space shall be permanently marked as "For Future 240V use"; and • Either a dedicated cold water supply, or the cold water supply shall pass through the designated HPWH location just before reaching the gas or propane water heater; and • The hot water supply pipe coming out of the gas or propane water heater shall be routed first through the designated HPWH location before serving any fixtures; and • The hot and cold water piping at the designated HPWH location shall be exposed and readily accessible for future installation of a HPWH; and

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	A condensate drain no more than 2 inches higher than the base of the installed water heater, and allows natural draining without pump assistance.	
05	For Air-Source Heat Pump Water Heaters (HPWH), the following shall be met (Section 110.3(c)7) a) Backup heat is required when inlet air is unconditioned, unless the compressor cutout cut-off temperature is below the Heating Winter Median of Extreme. Backup heat may be internal or external to the HPWH b) Meet one of the ventilation requirements below. Minimum volume and opening size requirements shall be the sum of all HPWHs installed within the same space. Compressor capacity shall be determined using AHRI 540 Table 4 reference conditions for refrigeration with the "High" rating test point: a. Installed using a method provided by the manufacturer to meet or exceed the level of performance provided by the ventilation requirements of Section 110.3(c)7B2 through Section 110.3(c)B4. b. For HPWH installation without ducts, the installation space shall have a volume not less than the greater of 100 cubic feet per kBtu per hour of compressor capacity, or the minimum volume provided by the manufacturer for this method; or c. For HPWH installation without ducts, the installation space shall be vented to a communicating space via permanent openings, according to the following requirements: i. Communicating space shall meet the minimum volume of Section 110.3(c)7B12 above, minus the volume of the HPWH installation space; and ii. Permanent openings shall consist of a single layer of fixed flat slat louvers or grilles, with a total minimum Net Free Area (NFA) the larger of 125 square inches plus 25 square inches per kBtu per hour of compressor capacity, or the minimum provided by the manufacturer for this method. The permanent openings shall be fully louvered doors or two openings of equal area, one in the upper half of the enclosure and one in the bottom half of the enclosure. The top of the upper opening must be 12 inches or less from the enclosure top and the bottom of the lower vent must be 12 inches or less from the enclosure bottom; or d. For HPWH installations with ducts, the following requirements shall be met: i. The space joined to the installation space via ducts shall meet the minimum volume of Section 110.3(c)7B2 above, minus the volume of the HPWH installation space; and ii. All duct connections and building penetrations shall be sealed; and iii. Exhaust air ducts and all ducts which cross pressure boundaries shall be insulated to minimum of R-6; and iv. Where only the HPWH inlet or outlet is ducted, installation space shall include permanent openings which consist of a single layer of fixed flat slat louvers or grilles in the bottom half of the room, and/or a door undercut. With a ducted inlet, the minimum NFA shall be equal to the cross-sectional area of the duct. With a ducted exhaust, the minimum NFA shall be the larger of 20 square inches or the minimum NFA provided by the manufacturer for this method; and v. Where the inlet and outlet ducts both terminate within the same pressure boundary, airflow from the termination points shall be diverted away from each other.	
05	Verification Status:	☐ Pass - all applicable requirements are met; or ☐ Fail - one or more applicable requirements are not met. Enter reason for failure in corrections notes field below; or ☐ All N/A - This entire table is not applicable
06	Correction Notes:	

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G. ~~HERS~~-Verified Compact Hot Water Distribution Expanded Credit (CHWDS-H-EX) (RA3.6.5)

For dwelling units with multiple systems, enter the master bath distance and kitchen distance to the closest water heater, and enter the average of the furthest fixture to each water heater.

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met unless otherwise noted in the Verification Status and the Corrections Notes in this table.

01	02	03	04	05	06	07	08	09
Dwelling Name	Number of Stories	Master Bath distance of furthest fixture to Water Heater in feet	Kitchen distance from furthest fixture to Water Heater in feet	Furthest Third furthest fixture to Water Heater in feet (Avg for multiple water heaters)	Weighted Distance	Qualification Distance	Design Compactness Factor	Calculated Compactness Factor
08	No hot water piping >1 inch diameter is allowed.							
09	Length of 1 inch diameter piping is limited to 8 feet or less.							
10	Two and three story buildings cannot have hot water distribution piping in the attic, unless the water heater is also located in the attic.							
11	Eligible recirculating systems must be HERS -Verified Demand Recirculation: Manual Control conforming to RA4.4.17.							
12	Verification Status:	☐ Pass - all applicable requirements are met; or ☐ Fail - one or more applicable requirements are not met. Enter reason for failure in corrections notes field below; or ☐ All N/A - This entire table is not applicable						
13	Correction Notes:							

H. Compact Hot Water Distribution (~~CHWDS~~) (RA4.4.6)

For dwelling units with multiple systems, enter the master bath distance and kitchen distance to the closest water heater, and enter the average of the furthest fixture to each water heater.

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met

01	02	03	04	05	06	07	08	09
Dwelling Name	Number of Stories	Master Bath distance of furthest fixture to Water Heater in feet	Kitchen distance from furthest fixture to Water Heater in feet	Furthest Third furthest fixture to Water Heater in feet (Avg for multiple water heaters)	Weighted Distance	Qualification Distance	Design Compactness Factor	Calculated Compactness Factor

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I. ~~HERS~~-Verified Drain Water Heat Recovery System (DWHR-H) (RA3.6.9)

DWHR devices shall comply with these requirements

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met unless otherwise noted in the Verification Status and the Corrections Notes in this table.

Design DWHR System Information

01	02	03	04
System ID/Name	Rated Effectiveness	Installation Configuration	Percent of shower served by the DWHR device

Installed DWHR System Information

05	06	07	08	09	10	11
System Name/ID	Manufacturer	Model #	Rated effectiveness	Installation Configuration	Percent of shower served by the DWHR device	DWHR System Certified by CEC (Yes/No)
12	For water heating system serving a single dwelling, the DWHR system shall, at the minimum, recover heat from the master bathroom shower and must at least transfer that heat either back to the respective shower(s) or the water heater.					
13	For central water heating system serving multiple dwellings, the DWHR system shall, at the minimum, recover heat from half the showers located above the first floor and must at least transfer that heat either back to all the respective showers or the water heater.					
14	The DWHR unit(s) shall be installed within 1 degrees of the rated slope. Sloped DWHR shall have a minimum lengthwise slope of 1 degree. The lateral level tolerance shall be within plus or minus 1 degree.					
15	Verification Status:			☐ Pass - all applicable requirements are met; or ☐ Fail - one or more applicable requirements are not met. Enter reason for failure in corrections notes field below; or ☐ All N/A - This entire table is not applicable		
16	Correction Notes:					

J. ~~HERS~~-Verified Pipe Insulation ~~Credit for Single Dwelling Systems~~-Requirements (~~PIC-H~~) (RA3.6.23)

Systems that utilize this distribution type shall comply with these requirements.

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met unless otherwise noted in the Verification Status and the Corrections Notes in this table.

01	HERSECC rater shall perform a visual inspection that all hot water piping comply with the insulation requirements in 150.0(1).		
02	Verification Status:	☐	Pass - all applicable requirements are met; or
		☐	Fail - one or more applicable requirements are not met. Enter reason for failure in corrections notes field below; or
		☐	All N/A - This entire table is not applicable
03	Correction Notes:		

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K. ~~HERS~~-Verified Central Parallel Piping Requirements (PP-H) (RA3.6.4)

Systems that utilize this distribution type shall comply with these requirements.

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met unless otherwise noted in the Verification Status and the Corrections Notes in this table.

01	Each central manifold has 5 feet or less of pipe between manifold and water heater.	
02	For manifolds that include valves, the manifold must be readily accessible in accordance with the plumbing code.	
03	Hot water distribution system piping from the manifold to the fixtures and appliances must take the most direct path. For example, piping from a second story manifold cannot supply the first floor.	
04	The hot water distribution piping must be separated by at least 2 inches from any other hot water supply piping, and at least 6 inches from any cold water supply piping. Alternatively, the hot water supply piping must be insulated to the thicknesses shown in TABLE 120.3-A-1.	
05	Verification Status:	☐ Pass - all applicable requirements are met; or ☐ Fail - one or more applicable requirements are not met. Enter reason for failure in corrections notes field below; or ☐ All N/A - This entire table is not applicable
06	Correction Notes:	

L. Central Parallel Piping Requirements (~~PP~~) (RA4.4.4)

Systems that utilize this distribution type shall comply with these requirements.

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met.

01	Each central manifold has 15 feet or less of pipe between manifold and water heater	
02	For manifolds that include valves, the manifold must be readily accessible in accordance with the plumbing code.	
03	Hot water distribution system piping from the manifold to the fixtures and appliances must take the most direct path. For instance, piping from a second story manifold cannot supply the first floor	
04	The hot water distribution piping must be separated by at least 2 inches from any other hot water supply piping, and at least 6 inches from any cold water supply piping. Alternatively, the hot water supply piping must be insulated to the thicknesses shown in TABLE 120.3-A-1.	

M. Point of Use Requirements (POU) (RA4.4.5)

Systems that utilize this distribution type shall comply with these requirements.

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met.

01	All hot water supply pipe run lengths are equal to or less than the maximum values shown below, based on the pipe diameter. If a combination of piping is used in a single run, then one half the allowed length of each size is the maximum installed length. The maximum allowed length of piping for the longest run terminating in: 3/8 inch - For only one pipe size - max length allowed is 15 feet For combination pipe sizes the max allowed length of 3/8-inch piping is 7.5 feet, of 1/2 inch piping is 5 feet, and 3/4 inch piping is 2.5 feet. 1/2 inch - For only one pipe size – max length allowed is 10 feet For combination pipe sizes the allowed length of 1/2-inch piping is 5 feet, and 3/4 inch piping is 2.5 feet. 3/4 inch - For only one pipe size = 5 feet	
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N. Mandatory Requirements for all Recirculation Systems (RA4.4.7)

Systems that utilize this distribution type shall comply with these requirements.

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met

01	A check valve located between the recirculation pump and the water heater to prevent unintentional recirculation.
02	Piping must take most direct path between water heater and fixtures.
03	Insulation is not required on the cold water line when it is used as the return.
04	If more than one loop installed each loop shall have its own pump and controls.

O. Recirculation Non-Demand Controls Requirements ~~(R-ND)~~ (RA4.4.8)

Systems that utilize this distribution type shall comply with these requirements.

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met.

01	The active control shall be either: timer, temperature, or time and temperature. Timers shall be set to less than 24 hours. The temperature sensor shall be connected to the piping and to the controls for the pump.
02	If more than one loop is installed each loop shall have its own pump and controls.

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P. Demand Recirculation Manual Control (R-DRmc) (RA4.4.9)/Sensor Control (RDRsc) (RA4.4.10) Requirements

Systems that utilize this distribution type shall comply with these requirements.

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met.

01	The system operates "on-demand", meaning that the pump begins to operate shortly before or immediately after hot water draw begins, and stops when the return water temperature reaches a certain threshold value. For Demand Recirculation Manual Control, the pump shall be turned on using a manual switch system. For Demand Recirculation Sensor Control, the pump shall be turned on using a sensor system.
02	The controls shall be located in the kitchen, bathroom, and any hot water fixture location that is at least 20 feet from the water heater.
03	Manual controls may be activated by wired or wireless mechanisms. Each control shall have standby power of 1 Watt or less.
04	Sensor Controls may be activated by wired or wireless mechanisms, including buttons, motion sensors, door switches and flow switches. Each control shall have standby power of 1 Watt or less.
05	Pump and control placement shall meet one of the following criteria: - When a dedicated return line has been installed the pump, controls and thermo-sensor are installed at the end of the supply portion of the recirculation loop; or - The pump and controls are installed on the dedicated return line near the water heater and the thermo-sensor is installed in an accessible location as close to the end of the supply portion of the recirculation loop as possible; or - When the cold water line is used as the return, the pump, demand controls and thermo-sensor shall be installed in an accessible location at the end of supply portion of the hot water distribution line (typically under a sink).
06	After the pump has been activated, the controls shall allow the pump to operate until the water temperature at the thermo-sensor rises to one of the following values: - Not more than 10°F (5.6°C) above the initial temperature of the water in the pipe - Not more than 102°F (38.9°C).
07	Controls shall limit operation to no more than 5 minutes following activation.

Q. HERS-Verified Demand Recirculation Manual Control (RDRmc-H) (RA3.6.6)/Sensor Control (RDRsc-H) (RA3.6.7) Requirements

Systems that utilize this distribution type shall comply with these requirements

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met unless otherwise noted in the Verification Status and the Corrections Notes in this table.

01	HERSECC rater shall perform a visual inspection to verify that the demand pump, manual/sensor controls and thermo-sensor are present and operating properly consistent with the applicable requirements of RA4.4.9 and RA4.4.10	
02	Verification Status:	☐ Pass - all applicable requirements are met; or ☐ Fail - one or more applicable requirements are not met. Enter reason for failure in corrections notes field below; or ☐ All N/A - This entire table is not applicable
03	Correction Notes:	

R. Determination of HERS Verification Compliance

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DOCUMENTATION AUTHOR'S DECLARATION STATEMENT

1. I certify that this Certificate of Verification documentation is accurate and complete.

Documentation Author Name:	Documentation Author Signature:
Company:	Date Signed:
Address:	CEA/ <u>HERSECC</u> Certification Information (if applicable):
City/State/Zip:	Phone:

RESPONSIBLE PERSON'S DECLARATION STATEMENT

I certify the following under penalty of perjury, under the laws of the State of California:

1. The information provided on this Certificate of Verification is true and correct.
2. I am the certified HERSECC Rater who performed the verification identified and reported on this Certificate of Verification (responsible rater).
3. The installed features, materials, components, manufactured devices, or system performance diagnostic results that require HERSECC verification identified on this Certificate of Verification comply with the applicable requirements in Reference Appendices RA2, RA3, and the requirements specified on the Certificate of Compliance for the building approved by the enforcement agency.
4. The information reported on applicable sections of the Certificate(s) of Installation (CF2R) signed and submitted by the person(s) responsible for the construction or installation conforms to the requirements specified on the Certificate(s) of Compliance (CF1R) approved by the enforcement agency.
5. I understand that a registered copy of this Certificate of Verification shall be posted, or made available with the building permit(s) issued for the building, and made available to the enforcement agency for all applicable inspections, and I will take the necessary steps to ensure this requirement is accomplished.
6. I understand that a registered copy of this Certificate of Verification is required to be included with the documentation the builder provides to the building owner at occupancy, and I will take the necessary steps to ensure this requirement is accomplished.

BUILDER OR INSTALLER INFORMATION AS SHOWN ON THE CERTIFICATE OF INSTALLATION

Company Name (Installing Subcontractor, General Contractor, or Builder/Owner):	
Responsible Builder or Installer Name:	CSLB License:

HERSECC PROVIDER DATA REGISTRY INFORMATION

Sample Group Number (if applicable):	Dwelling Test Status in Sample Group (if applicable):
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HERSECC RATER INFORMATION

<u>HERSECC</u> Rater Company Name:	
Responsible Rater Name:	Responsible Rater Signature:
Responsible Rater Certification Number w/ this <u>HERSECC</u>	Date Signed:

Registration Number:

Registration Date/Time:

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For information and data collection
only. Not valid until registered with an
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CF3R-PLB-22-H User Instructions

A. Design ~~HERS~~ Verified Central Water Heating Systems Information

This table reports the water heating system features that were specified on the registered CF1R compliance document for this project. For information only and requires no user input.

A2. Design ~~HERS~~ Verified Dwelling Unit HPWH System Information

This table reports the water heating system features that were specified on the registered CF1R compliance document for this project. This section is for information/verification purposes only and requires no user input.

B. Installed ~~HERS~~ Verified Dwelling Unit Water Heating Systems Information

This table reports the water heating system information that is being installed. Require one line for each system.

1. Dwelling Unit Name - Reference information from CF1R.
2. Water Heating System ID or Name – Reference information from CF1R.
3. Water Heating System Type – Reference information from CF1R. The different kinds of water heating system type are DHW, or Combined Hydronic.
4. Water Heater Type – Information from CF1R. The different kinds of water heaters are Large/Commercial Storage, Small/Consumer Storage, Residential-Duty Commercial Storage, Heat Pump, Boiler, Large/Commercial Instantaneous, Small/Consumer Instantaneous, Residential-Duty Commercial Instantaneous or Indirect.
5. # of Like (or Identical) Water Heaters in system – Reference information from CF1R.
6. Fuel Type – Reference information from CF1R. The different kinds of fuel types are natural gas, propane, oil, or electricity.
7. Rated Input Type – Reference information from CF1R. For natural gas, propane and oil fuel type the input type is Btu/hr. For electric the input type is kW.
8. Rated Input Value – User input. Numerical value of the rated input. Must be equal to or less than value indicated on the CF1R.
10. Dwelling Unit DHW System Distribution Type - Reference information from CF1R.
11. Compact Distribution - Reference information from CF1R.
12. Drain Water Heat Recovery - Reference information from CF1R.

B2. Installed ~~HERS~~ Verified Dwelling Unit HPWH System Information

This table reports the water heating system information that is being installed. Require one line for each installed water heater. Not applicable for central systems.

1. Dwelling Unit Name – Reference information from Table C.

2. Water Heating System ID or Name – Reference information from Table C. AFUE, UEF and Thermal Efficiency.
3. Modeled Equipment Make and Model – User input must be equal to the value indicated on Table C as default and allow user to override with an equivalent system based on the simulated equipment in Table C.
4. # of Like (or Identical) Water Heaters in System – Reference information from Table C.
5. Tank Location – User input. Must be equal to value indicated in Table C.
6. Exterior Tank Insulation R-value – User input. Must be equal to or higher than value indicated in Table C.
- 6a. Tank Volume – User input must equal reference information on Table A2.
7. Dwelling Unit DHW System Distribution Type – Reference information from Table C.
8. Compact Distribution – Reference information from Table C.

C. Design ~~HERS~~ Verified Dwelling Unit Water Heating Efficiency Information

This table reports the water heating system features that were specified on the registered CF1R compliance document for this project. For information only and requires no user input.

D. Installed ~~HERS~~ Verified Dwelling Unit Water Heating Efficiency Information

This table reports the water heating system information that is being installed. Require one line for each central system.

1. Water Heating System ID or Name – Reference information from CF1R02 Heating Efficiency Type – Reference information from CF1R. Different efficiency types are Energy Factor, AFUE, UEF and Thermal Efficiency.
2. Heating Efficiency Value – User input. Numerical value of the Heating Efficiency. Must be equal to or higher efficiency than value indicated on the CF1R.
3. Standby Loss – User input. Must be equal to or less than value indicated on the CF1R. Value may be N/A if CF1R value is N/A.
4. Exterior Insulation R-Value – User input. Must be equal to or higher than value indicated on the CF1R. Value may be N/A if CF1R value is N/A.
5. Water Heater Storage Volume (gal) – User input. Value may be N/A if water heater type is instantaneous with zero storage.
6. Tank location – User input. Must be equal to system type indicated on the CF1R.

E. Installed Water Heater Manufacturer Information

This table reports the manufacturer information of the installed water heater(s). Require one line for each installed water heater

1. Water Heating System ID or Name – Reference information from CF1R.
2. Manufacturer – User input. Enter the name of the water heater manufacturer.
3. Model Number – User input. Enter the model number of the water heater.

F. Mandatory Measures for ~~all Domestic Hot Water Distribution~~ Single Dwelling Systems

This table lists the requirements for ~~all DHW~~ Single Dwelling systems. ~~HERS~~ ECC rater must ensure all the requirements in this table are met.

G. ~~HERS~~-Verified Compact Hot Water Distribution Expanded Credit and

H. Compact Hot Water Distribution Basic

If performance compliance is used, this table lists the values used in the performance calculation and require no user input.

If prescriptive compliance is used, fill out this table

1. Reference information from CF1R
2. Enter the Master Bath distance of furthest fixture to Water Heater in feet. For multiple water heaters, enter the distance to the closest water heater.
3. Enter the Kitchen distance from furthest fixture to Water Heater in feet. For multiple water heaters, enter the distance to the closest water heater.
4. Enter Furthest Third fixtures from fixture to Water Heater in feet. For multiple water heaters, enter the average of the furthest distance of each water heater.
5. Calculated value – no user input required
6. Calculated value – no user input required

I. ~~HERS~~-Verified Drain Water Heat Recovery System

This table lists the requirements for all central recirculation systems. ~~HERS~~ECC rater must ensure all the requirements in this table are met.

1. Reference information from CF1R.
2. Reference information from CF1R.
3. Reference information from CF1R.
4. Reference information from CF1R.
5. Reference information from CF1R.
6. Drain Water Heat Recovery Manufacturer's name- Enter the name of the Manufacturer.
7. Drain Water Heat Recovery Manufacturer's model number – Enter the Model number.
8. Rated Effectiveness' – Enter the rated effectiveness of the DWHR device.
9. Installation Configuration – Enter type of configuration. Available options are Equal flow, unequal to shower, and unequal to water heater
10. Percent of shower served by the DWHR device – Enter the percent of showers served by this DWHR device.
11. DWHR System Certified by CEC – Enter "Yes" if certified or else enter "No".

J. ~~HERS~~-Verified Pipe Insulation for Central Systems~~Credit~~ Requirements

This table only applies to systems indicated as ~~HERS~~-Verified Pipe Insulation Credit. In addition to the mandatory requirements in Table F, the ~~HERS~~ECC rater must ensure the requirements in this table are met.

K. ~~HERS~~-Verified Central Parallel Piping Requirements

This table only applies to systems indicated as ~~HERS~~-Verified Central Parallel Piping. In addition to the mandatory requirements in Table F, the ~~HERS~~ECC rater must ensure the requirements in this table are met.

L. Central Parallel Piping Requirements

This table only applies to systems indicated as **Central Parallel Piping**. In addition to the mandatory requirements in Table F, the installer must ensure the requirements in this table are met.

M. Point of Use Requirements

This table only applies to systems indicated as **Point of Use**. In addition to the mandatory requirements in Table F, the installer must ensure the requirements in this table are met.

N. Mandatory Requirements for all Recirculation Systems

The requirements of this table apply to all recirculation systems listed below.

O. Recirculation Non-Demand Controls Requirements

This table only applies to systems indicated as **Recirculation Non-demand controls**. In addition to the mandatory requirements in Table F and N, the installer must ensure the requirements in this table are met.

P. Demand Recirculation Manual Control/Sensor Control Requirements

This table only applies to systems indicated as **Demand Recirculation Manual Control, Demand Recirculation Sensor Control, ~~HERS~~-Verified Demand Recirculation Manual Control or ~~HERS~~-Verified Demand Recirculation Sensor Control**. In addition to the mandatory requirements in Table F and N, the installer must ensure the requirements in this table are met.

Q. ~~HERS~~-Verified Demand Recirculation Manual Control (RDRmc-H) (RA3.6.6)/Sensor Control (RDRsc-H) (RA3.6.7)

This table only applies to systems indicated as **~~HERSECC~~-Verified Demand Recirculation Manual Control or ~~HERSECC~~-Verified Demand Recirculation Sensor Control**. In addition to the mandatory requirements in Table F and N, the ~~HERS~~ rater/installer must ensure the requirements in this table are met.

A. Design **HERS-Verified Dwelling Unit Water Heating Systems Information (other than HPWH)**

This table reports features of the water heating system other than **HPWH** systems that were specified on the registered CF1R compliance document for this project.

<< if CF1R do not have any non-HPWH systems, then display the "section does not apply" message; else require one row of data for each non-HPWH Water Heating System name identified on the CF1R report>>

01	02	03	04	05	06	07	08	10	11	12
Dwelling Unit Name	Water Heating System ID or Name	Water Heating System Type	Water Heater Type	# of Like (or Identical) Water Heaters in System	Fuel Type	Rated Input Type	Rated Input Value	Dwelling Unit DHW System Distribution Type	Compact Distrib.	Drain Water Heat Recovery
<<reference value from CF1R; if Single Family, then value = Single Family>>	<<reference values from CF1R (see rule in header)>>	<<reference values from CF1R allowed values= DHW, Combined Hydronic, Hydronic>>	<<reference values from CF1R Allowed values = Boiler, Indirect, Consumer Instantaneous, Commercial Instantaneous, Consumer Storage, Commercial Storage, Residential -Duty Commercial Storage, or Residential -Duty Commercial Instantaneous >>	<<reference values from CF1R>>	<<reference values from CF1R. Allowed values are Heat Pump, Electric Resistance, Natural Gas, or Propane >>	<<If A06 = Heat Pump, then value = NA; Else reference values from CF1R. Allowed values: If Fuel Type A06 = Natural Gas, Propane then value = Btu/Hr. Else if Fuel Type = Electric Resistance then value = kW >>	<<if A06 = Heat Pump, then result = NA; If performance, reference values from CF1R-PRF; Elseif Prescriptive, then value = NA >	<<reference values from CF1R If performance and A09 = NA then allowed values are *Standard Distribution System * Point of Use *Central Parallel Piping *Recirculation System Non-Demand Control * Demand Recirculation Manual Control * Demand Recirculation Sensor Control * HERS -Verified Pipe Insulation for Central Systems Credit; * HERS -Verified Parallel Piping * HERS -Verified Demand Recirculation Manual Control * HERS -Verified Demand Recirculation Sensor Control; Else if A09 ≠ NA then allowed values are * Standard Distribution System * HERS -Verified Pipe Insulation for Central Systems Credit; Else if prescriptive, Allowed values are *Standard Distribution System *Demand Recirculation * Demand Recirculation Manual Control>>	<< If parent CF1R = NCB-01, then if NCB-01 Field M09 = CompactH otWaterDistributionSystem HERSECC or CompactH otWaterDistributionSystemBasic or CompactDistributionDrainHeatRecovery HERSECC then value is BasicType else NotCompact; Else if parent CF1R = ALT-01 or ADD-01, then value = NotCompact; Else if information not available from PRF-01, allow user select. Allowed values are *Basic *Expanded *NotCompact; Else use value from PRF-01; >>	<<If parent CF1R = NCB-01, then if NCB-01 Field M09 = CompactDistributionDrainHeatRecovery HERSECC or DrainWaterHeatRecovery HERSECC then value is DWHR else NoDWHR; Else if parent CF1R = ALT-01 or ADD-01, then value = NoDWHR; Else if information not available from PRF-01, allow user select. Allowed values are *DWHR *NoDWHR; Else use value from PRF-01; >>

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A2. Design ~~HERS~~ Verified Dwelling Unit HPWH System Information

This table reports the water heating system(s) that were specified on the registered CF1R compliance document for this project.

<<If CF1R has HPWH, then require table, else display the "section does not apply" message>>

<<require one row of data for each HPWH water heater identified on the CF1R-PRF>>

01	02	03	04	05	06	06a	07	08	09	10
Dwelling Unit Name	Water Heating System ID or Name	Modeled Equipment Make and Model	# of Like (or Identical) Water Heaters in System	Tank Location	Exterior Tank Insulation R-value	Tank Volume	Dwelling Unit DHW System Distribution Type	Compact Distribution.	Drain Water Heat Recovery	Simulated Equipment Make and Model
<<reference value from CF1R; if Single Family, then value = Single Family>>	<<reference value from CF1R>>	<<reference value from CF1R-PRF. Else if prescriptive, display "NEEA Tier 3" >>	<<reference values from CF1R>>	<<Reference value from CF1R>>	<<reference values from CF1R-PRF; else NA>>	<<reference values from CF1R-PRF; else if parent CF1R = NCB-01 or ADD-01, then allow user input; else NA >>	<<reference values from CF1R If performance, then allowed values are *Standard Distribution System * Point of Use * Central Parallel Piping * Recirculation System Non-Demand Control * Demand Recirculation Manual Control * Demand Recirculation Sensor Control * HERS Verified Pipe Insulation for Central Systems Credit * HERS Verified Parallel Piping * HERS Verified Demand Recirculation Manual Control * HERS Verified Demand Recirculation Sensor Control; Else if prescriptive, Allowed values are *Standard Distribution System	<<If parent CF1R = NCB-01, then if NCB-01 Field M09 = CompactHotWaterDistributionSystem HERSECC or CompactHotWaterDistributionSystemBasic or CompactDistributionDrainHeatRecovery HERSECC then value is BasicType else NotCompact; Else if parent CF1R = ALT-01 or ADD-01, then value = NotCompact; Else if information not available from PRF-01, allow user select. Allowed values are *Basic *Expanded *NotCompact;	<<If parent CF1R = NCB-01, then if NCB-01 Field M09 = CompactDistributionDrainHeatRecovery HERSECC or DrainWaterHeatRecovery HERSECC then value is DWHR else NoDWHR; Else if parent CF1R = ALT-01 or ADD-01, then value = NoDWHR; Else if information not available from PRF-01, allow user select. Allowed values are *DWHR *NoDWHR; Else use value from PRF-01; >>	<<if performance, hide column from user, needed for equivalency lookup; Reference value from XML; Elseif prescriptive, do not require field>>

								*Demand Recirculation * Demand Recirculation Manual Control>>	Else use value from PRF-01; >>		
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B. Installed ~~HERS~~ Verified Dwelling Unit Water Heating Systems Information

This table reports features of the water heating system other than HPWH systems installed in this project.

<< if Table A does not apply, then display the "section does not apply" message; else require one row of data for each Dwelling Unit Water Heating System name identified in table A>>

01	02	03	04	05	06	07	08	10	11	12
Dwelling Unit Name	Water Heating System ID or Name	Water Heating System Type	Water Heater Type	# of Like (or Identical) Water Heaters in System	Fuel Type	Rated Input Type	Rated Input Value	Dwelling Unit DHW System Distribution Type	Compact Distrib.	Drain Water Heat Recovery
<<Reference value from A01>>	<<Reference value from A02>>	<<reference value from A03>>	<<reference value from A04>>	<<reference value from A05>>	<<Reference value from A06>>	<<Reference value from A07>>	<<User input value which must pass the following range tests: Value may be NA if B08 is NA. If B06 = Natural Gas or Propane, then If B04 = Commercial Storage, then value must be > 75,000 Btu/hr; If B04 = Consumer Storage, then value must be ≤ 75,000 Btu/hr; If B04 = Commercial Instant, then value must be > 200,000 Btu/hr; If B04 = Consumer Instant, then value must be ≤ 200,000 Btu/hr;	<<Reference value from Design Dwelling Unit DHW System Distribution A10>>	<<Reference value from A11>>	<<Reference value from A12>>

							Else if B04 = Residential I-Duty Commercial Storage, then value must be ≤ 105,000 Btu/hr; Else if B06 = Electricity, then If B04 = Commercial Storage or Commercial Instant, then value must be > 12 kW; If B04 = Consumer Storage or Consumer Instant, then value must be ≤ 12 kW; Else if B04 = Residential I-Duty Commercial Instantaneous, then value must be ≤ 58.6 kW; If the value passes range test and if A06 = Electric Resistance, it is stored in WaterHeaterElectric FiredRate Input, Otherwise the value			
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							is stored in WaterHeaterGasFire dRatedInput>>			

B2. Installed ~~HERS~~ Verified Dwelling Unit HPWH System Information

This table reports the water heating system(s) that were installed in this project.

<<If Table A2 does not apply, then require table, else display the "section does not apply" message>>

<<require one row of data for each water heater identified in table A2>>

01	02	03	04	05	06	06a	07	08	09
Dwelling Unit Name	Water Heating System ID or Name	Modeled Equipment Make and Model	# of Like (or Identical) Water Heaters in System	Tank Location	Exterior Tank Insulation R-value	Tank Volume	Dwelling Unit DHW System Distributio n Type	Compact Distrib.	Drain Water Heat Recovery
<<Reference value from A2-01 >>	<<reference value from A2-02>>	<<if performance , user input is equal to C03 as default, and allow user to override with an equivalent system based on the simulated equipment in A2-10; elseif prescriptive newly constructed, If CF1R M03 = 1 and # of bedroom >1, then allow user to enter any 240V HPWH; Else if CF1R M03=1 and # of bedroom <= 1, then allow user to enter any HPWH. Elseif M03=2 , allow user to enter any Tier 3 Or higher model>>	<<Reference value from A2-04>>	<<Reference value from A2-05.>>	<<User input value; check value must be ≥A2-06 to comply, else flag non- compliant values and do not allow the doc to be registered. Value may be NA if A2- 06 is NA >>	<<User input value; check value must be ≥A2-06a to comply, else flag non- compliant values and do not allow the doc to be registered. Value may be NA if A2- 06a is NA >>	<< Reference values from A2-07>>	<< Reference values from A2-08>>	<< Reference values from A2-09>>
10	Compliance Statement:		<<calculated field: If A2-06 ≤ B2-06, then display result = System complies; else display result = system does not comply>>						

C. Design **HERSVerified Dwelling Unit Water Heating Efficiency Information**

This table reports the water heater(s) efficiency features specified on the registered CF1R compliance document for this project.

<< require one row of data for each water heater identified in Section A and A2 >>

01	02	03	04	05	06	07
Water Heating System ID or Name	Heating Efficiency Type	Heating Efficiency Value	Standby Loss (%)	Exterior Insulation R-Value	Water Heater Storage Volume (gal)	Tank location
<<Reference value from design Water Heating System ID or Name (A02) or A2-02>>	<< reference values from CF1R; if parent CF1R = CF1R-ADD, ALT or NCB, then value = NA; else if parent = CF1R-PRF, then allowed values are: *Uniform Energy Factor *AFUE *Thermal Efficiency >>	<< reference values from CF1R-PRF-01; Else = NA>>	<<reference values from CF1R-PRF-01. Else = NA >>	<<reference values from CF1R-PRF-01. Else = NA >>	<<reference values from CF1R. NA is allowed only if Water Heater Type = Consumer Instantaneous, Commercial Instantaneous, or Residential-Duty Commercial Instantaneous >>	<<Reference value from CF1R>>

D. Installed **HERSVerified Dwelling Unit Water Heating Efficiency Information**

This table reports the water heating system efficiency features that were installed in this project.

<<require one row of data for each water heart identified in Section C >>

01	02	03	04	05	06	07
Water Heating System ID or Name	Heating Efficiency Type	Heating Efficiency Value	Standby Loss (%)	Exterior Insul. R-Value	Water Heater Storage Volume (gal)	Tank location
<<Reference value from design Water Heating System ID or Name (C01)>>	<< If C02 = N/A then allow user to select from list: *Uniform Energy Factor, *AFUE, or *Thermal Efficiency; else reference value from C02 as default>>	<<User input value; check value must be ≥ value in E03 to comply, else flag non-compliant values and do not allow the doc to be registered, . Value may be NA if C03 is NA >>	<<User input value; check value must be ≤ value in C04 to comply, else flag non-compliant values and do not allow the doc to be registered. Value may be NA if C04 is NA >>	<<User input value; check value must be ≥ value in E05 to comply, else flag non-compliant values and do not allow the doc to be registered. Value may be NA if C05 is NA >>	<< User Input must equal reference values from E06, else flag non-compliant values and do not allow the doc to be registered. NA is allowed only if Water Heater Type = Consumer Instantaneous or Commercial Instantaneous >>	<<Reference value from C07 Value may be NA if CF1R value is NA >>
08	Compliance Statement	<<calculated field: If D03 = NA or D03 ≥ C03, D04 = NA or D04 ≤ C04, D05 = NA or D05 ≥ C05, D06 = C06, and D07 = C07, then display result = System complies; else display result = system does not comply>>				

E. Installed Water Heater Manufacturer Information

<<require one row of data for each water heater identified in Section C >>

01	02	03
Water Heating System ID or Name	Manufacturer	Model Number
<<reference value from B02 or B2-02>>	<<User input>>	<<User input>>

F. Mandatory Measures for ~~all Domestic Hot Water Distribution~~Single Dwelling Systems

01	Equipment shall meet the applicable requirements of the Appliance Efficiency Regulations (Section 110.3(b)1).
02	Unfired Storage Tanks are insulated with an external R-3.5 or combination of R-16 internal and external Insulation. (Section 110.3(c)34).
03	<u>Pipe Insulation (Section 150(J))</u> • All domestic hot water piping shall be insulated as specified in Section 609.12 of the California Plumbing Code. Insulation buried below grade must be installed in a waterproof and non-crushable casing or sleeve. • Pipe insulation shall fit tightly and all elbows and tees shall be fully insulated. • Piping that penetrates framing members shall not be required to have pipe insulation for the distance of the framing penetration. • Piping that penetrates metal framing shall use grommets, plugs, wrapping or other insulating material to assure that no contact is made with the metal framing. Insulation shall butt securely against all framing members. • Piping surrounded with a minimum of 1 inch of wall insulation, 2 inches of crawlspace insulation, or 4 inches of attic insulation shall not be required to have pipe insulation. The following pipes are insulated, to the thicknesses required by Table 120.3A, except for those sections of pipe that are subject to one of the exceptions below: All domestic hot water piping shall be insulated as specified in Section 609.11 of the California Plumbing Code. Insulation buried below grade must be installed in a water proof and non-crushable casing or sleeve. • Piping from the heating source to storage tank or between tanks • Piping that penetrates framing members shall not be required to have pipe insulation for the distance of the framing penetration. Piping that penetrates metal framing shall use grommets, plugs, wrapping or other insulating material to assure that no contact is made with the metal framing. Insulation shall butt securely against all framing members. • Piping installed in interior or exterior walls that is surrounded on all sides by at least 1 inch (2.5 cm) of insulation. • Piping installed in crawlspace with a minimum of 1 inches (2.5 cm) of crawlspace insulation above and below. • Piping installed in attics with a minimum of 4 inches (10 cm) of attic insulation on top • Pipe insulation shall fit tightly and all elbows and tees shall be fully insulated.
04	For Gas or Propane Water Heaters: Ensure either a or b are installed (Section 150.0(n)) c) A designated space at least 2.5 feet by 2.5 feet and 7 feet tall within 3 feet from the water heater • A dedicated 125V, 20A electrical receptacle connected to the electric panel with a 120/240V 3 conductor, 10 AWG copper branch circuit <u>rated at 30 amps minimum</u>, within 3 feet from the water heater and accessible with no obstructions; • The conductor shall be labeled with the work "Spare" on both ends; and • A reserved single pole circuit breaker space next to the circuit breaker next to the branch circuit labeled "Future" 240V <u>use</u> shall be provided. • A condensate drain no more than 2 inches higher than the base of the water heater, <u>and allows for</u> natural draining <u>without pump assistance</u> d) A designated space at least 2.5 feet by 2.5 feet and 7 feet tall more than 3 feet from the water heater • A dedicated 240 volt branch circuit shall be installed within 3 feet from the designated space. The • branch circuit shall be rated at 30 amps minimum. The blank cover shall be identified as "240V ready"; and • The main electrical service panel shall have a reserved space to allow for the installation of a double pole circuit breaker for a future HPWH installation. The reserved space shall be permanently marked as "For Future 240V use"; and • Either a dedicated cold water supply, or the cold water supply shall pass through the designated HPWH location just before reaching the gas or propane water heater; and • The hot water supply pipe coming out of the gas or propane water heater shall be routed first through the designated HPWH location before serving any fixtures; and • The hot and cold water piping at the designated HPWH location shall be exposed and readily accessible for future installation of a HPWH; and • A condensate drain no more than 2 inches higher than the base of the installed water heater, and allows natural draining without pump assistance.

05	<u>For Air-Source Heat Pump Water Heaters (HPWH), the following shall be met (Section 110.3(c)7)</u> a) <u>Backup heat is required when inlet air is unconditioned, unless the compressor cutout cut-off temperature is below the Heating Winter Median of Extreme. Backup heat may be internal or external to the HPWH</u> b) <u>Meet one of the ventilation requirements below. Minimum volume and opening size requirements shall be the sum of all HPWHs installed within the same space. Compressor capacity shall be determined using AHRI 540 Table 4 reference conditions for refrigeration with the “High” rating test point:</u> a. <u>Installed using a method provided by the manufacturer to meet or exceed the level of performance provided by the ventilation requirements of Section 110.3(c)7B2 through Section 110.3(c)B4.</u> b. <u>For HPWH installation without ducts, the installation space shall have a volume not less than the greater of 100 cubic feet per kBtu per hour of compressor capacity, or the minimum volume provided by the manufacturer for this method; or</u> c. <u>For HPWH installation without ducts, the installation space shall be vented to a communicating space via permanent openings, according to the following requirements:</u> i. <u>Communicating space shall meet the minimum volume of Section 110.3(c)7B12 above, minus the volume of the HPWH installation space; and</u> ii. <u>Permanent openings shall consist of a single layer of fixed flat slat louvers or grilles, with a total minimum Net Free Area (NFA) the larger of 125 square inches plus 25 square inches per kBtu per hour of compressor capacity, or the minimum provided by the manufacturer for this method. The permanent openings shall be fully louvered doors or two openings of equal area, one in the upper half of the enclosure and one in the bottom half of the enclosure. The top of the upper opening must be 12 inches or less from the enclosure top and the bottom of the lower vent must be 12 inches or less from the enclosure bottom; or</u> d. <u>For HPWH installations with ducts, the following requirements shall be met:</u> i. <u>The space joined to the installation space via ducts shall meet the minimum volume of Section 110.3(c)7B2 above, minus the volume of the HPWH installation space; and</u> ii. <u>All duct connections and building penetrations shall be sealed; and</u> iii. <u>Exhaust air ducts and all ducts which cross pressure boundaries shall be insulated to minimum of R-6; and</u> iv. <u>Where only the HPWH inlet or outlet is ducted, installation space shall include permanent openings which consist of a single layer of fixed flat slat louvers or grilles in the bottom half of the room, and/or a door undercut. With a ducted inlet, the minimum NFA shall be equal to the cross-sectional area of the duct. With a ducted exhaust, the minimum NFA shall be the larger of 20 square inches or the minimum NFA provided by the manufacturer for this method; and</u> <u>Where the inlet and outlet ducts both terminate within the same pressure boundary, airflow from the termination points shall be diverted away from each other</u>	
	05	Verification Status: ☐ Pass - all applicable requirements are met; or ☐ Fail - one or more applicable requirements are not met. Enter reason for failure in corrections notes field below; or ☐ All N/A - This entire table is not applicable
06	Correction Notes: <<if Verification Status= Fail, then text entry in this Corrections Notes field is required; user input text>> The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met unless otherwise noted in the Verification Status and the Corrections Notes in this table.	

G. ~~HERS~~-Verified Compact Hot Water Distribution Expanded Credit (CHWDS-H-EX) (RA3.6.5)

For dwelling units with multiple systems, enter the master bath distance and kitchen distance to the closest water heater, and enter the average of the furthest fixture to each water heater.

<< Require one row of data, reporting the longest distances, for each dwelling identified in Table B or B2. with B11 or B2-08 = Expanded. If no dwelling in B11 or B2-08 = Expanded, then display section header and standard “This section does not apply” message>>

01	02	03	04	05	06	07	08	09
Dwelling Name	Number of Stories	Master Bath distance of furthest fixture to Water Heater in feet	Kitchen distance from furthest fixture to Water Heater in feet	Furthest Third furthest fixture to Water Heater in feet (Avg for multiple water heaters)	Weighted Distance	Qualification Distance	Design Compactness Factor	Calculated Compactness Factor
<<Reference value from	<<if performance, then value = NA;	<<Reference Value from CF1R-PRF;	<<Reference Value from CF1R-PRF;	<<Reference Value from CF1R-PRF;	<<Reference value from CF1R-PRF;	<< Reference Value from CF1R-PRF;	<<Reference value from CF1R-PRF;	<<Calculated value = 0.3 + 0.4 *

A01 or A2-01>>	Else if prescriptive, user select from list: 1, 2, 3>>	Else if prescriptive compliance or not in the XML, user input>>	Else if prescriptive compliance or not in the XML, user input>>	Else if prescriptive compliance or not in the XML, user input>>	else if prescriptive or not in the XML and A10 or A2-07 = Standard Distribution System, then value = (G03*0.4) +(G04*0.4) +(G05*0.2); else if A10 or A2-07 = Demand Recirculation Manual Control, then value = I05>>	Else if prescriptive compliance or not in the XML, value = ((a+b *CFA)/n) >> Where: a, b = Qualification distance coefficients from Table 4.4.6-2 below, CFA = Conditioned floor area of the dwelling unit (ft²) from CF1R, and n = Number of water heaters in the dwelling unit from A05 or A2-04 (unitless).	elseif prescriptive, then value = n/a>>	(G06/G07), if performance, value must be ≤ G08; elseif prescriptive, value must be ≤ 0.7 >>
08	No hot water piping >1 inch diameter is allowed.							
09	Length of 1 inch diameter piping is limited to 8 feet or less.							
10	Two and three story buildings cannot have hot water distribution piping in the attic, unless the water heater is also located in the attic.							
11	Eligible recirculating systems must be HERS -Verified Demand Recirculation: Manual Control conforming to RA4.4.17.							
12	Verification Status:	☐ Pass - all applicable requirements are met; or ☐ Fail - one or more applicable requirements are not met. Enter reason for failure in corrections notes field below; or ☐ All N/A - This entire table is not applicable						
13	Correction Notes: <<if Verification Status= Fail, then text entry in this Corrections Notes field is required; user input text>>							
The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met unless otherwise noted in the Verification Status and the Corrections Notes in this table.								

H. Compact Hot Water Distribution (~~CHWDS~~) (RA4.4.6)

For dwelling units with multiple systems, enter the master bath distance and kitchen distance to the closest water heater, and enter the average of the furthest fixture to each water heater.

<< Require one row of data, reporting the longest distances, for each dwelling identified in Section B or B2. with B11 or B2-08 = Basic. If no dwelling in B11 or B2-08 = Basic, then display section header and standard "This section does not apply" message>>

01	02	03	04	05	06	07	08	09
Dwelling Name	Number of Stories	Master Bath distance of furthest fixture to Water Heater in feet	Kitchen distance from furthest fixture to Water Heater in feet	Furthest Third furthest fixture to Water Heater in feet (Avg for multiple water heaters)	Weighted Distance	Qualification Distance	Design Compactness Factor	Calculated Compactness Factor

<<Reference value from A01 or A2-01>>	<<if performance, then value = NA; Else if prescriptive, user select from list: 1, 2, 3>>	<<Reference Value from CF1R-PRF; Else if prescriptive compliance or not in the XML, user input>>	<<Reference Value from CF1R-PRF; Else if prescriptive compliance or not in the XML, user input>>	<<Reference Value from CF1R-PRF; Else if prescriptive compliance or not in the XML, user input>>	<<Reference value from CF1R-PRF; else if prescriptive or not in the XML and A10 or A2-07 = Standard Distribution System, then value = (J03*0.4) +(J04*0.4) +(J05*0.2); else if A10 or A2-07 = Demand Recirculation Manual Control, then value = H05>>	<< Reference Value from CF1R-PRF; Else if prescriptive compliance or not in the XML, value = ((a+b *CFA)/n) >> Where: a, b = Qualification distance coefficients from Table 4.4.6-2 below, CFA = Conditioned floor area of the dwelling unit (ft ²) from CF1R, and n = Number of water heaters in the dwelling unit from A2-04 (unitless).	<<Reference value from CF1R-PRF; elseif prescriptive, then value = n/a>>	<<Calculated value = 0.3 + 0.4 * (H06/H07), if performance, value must be ≤ H08; elseif prescriptive, value must be ≤ 0.7 >>
The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met								

Table 4.4.6-2: Coefficients for the Qualification Distance Calculation

<< do not show table, only use for equation in G07 and H07>>

Building Type	Coefficient a		Coefficient b	
	Non-Recirculating	Recirculating	Non-Recirculating	Recirculating
Single Family	Use when distribution type (A10 Or A2-07) is *Standard Distribution System	Use when distribution type (A10 Or A2-07) is * Demand Recirculation Manual Control		
One story	10	22.7	0.0095	0.0099
Two story	15	11.5	0.0045	0.0095
Three story	10	0.5	0.0030	0.014

I. ~~HERS~~-Verified Drain Water Heat Recovery System (DWHR-H) (RA3.6.9)

<<require one row of data for each drain water heat recovery system identified in Section B or B2. with B12 or B2-09 = Yes. Else report section header and standard "This section does not apply" message>>

Design DWHR System Information

01	02	03	04
System ID/Name	Rated Effectiveness	Installation Configuration	Percent of shower served by the DWHR device
<<Reference value from	<<If performance, reference value from CF1R-PRF. If prescriptive, display "≥ 42%">>	<<If performance, reference value from CF1R-PRF. If prescriptive, display "NA">>	<<If performance, reference value from CF1R-PRF If prescriptive, display "NA">>

A02 or A2-02>>			
Installed DWHR System Information			
05	06	07	08
System ID/Name	Manufacturer	Model #	Rated effectiveness
<<Reference value from B02 or B2-02>>	<<User input>>	<<User input>>	<<User input, value ≥ Rated Effectiveness in I02, 0.42≤I03<1 >>
			Installation Configuration << If performance, reference value from I03. If prescriptive, user pick from list: • Equal flow • Unequal to shower • Unequal to water heater>>
			Percent of shower served by the DWHR device
			<< If performance, reference value from I04. If prescriptive, User input, 0<I06≤100 >>
			DWHR System Certified by CEC ☐ Yes ☐ No
12	For water heating system serving a single dwelling, the DWHR system shall, at the minimum, recover heat from the master bathroom shower and must at least transfer that heat either back to the respective shower(s) or the water heater.		
13	For central water heating system serving multiple dwellings, the DWHR system shall, at the minimum, recover heat from half the showers located above the first floor and must at least transfer that heat either back to all the respective showers or the water heater.		
14	The DWHR unit(s) shall be installed within 1 degrees of the rated slope. Sloped DWHR shall have a minimum lengthwise slope of 1 degree. The lateral level tolerance shall be within plus or minus 1 degree.		
15	Verification Status: ☐ Pass - all applicable requirements are met; or ☐ Fail - one or more applicable requirements are not met. Enter reason for failure in corrections notes field below; or ☐ All N/A - This entire table is not applicable		
16	Correction Notes: <<if Verification Status= Fail, then text entry in this Corrections Notes field is required; user input text>>		
The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met unless otherwise noted in the Verification Status and the Corrections Notes in this table.			

J. ~~HERS~~-Verified Pipe Insulation for Single Dwelling Systems for Central Systems Credit Requirements (PIC-H) (RA3.6.3)

Systems that utilize this distribution type shall comply with these requirements. <<If there are no systems in column A10 or A2-07 that have a value = "~~HERS~~-Verified Pipe Insulation for Single Dwelling Systems Credit", then display the "section does not apply" message; else display this entire table >>

01	HERS -ECC rater shall perform a visual inspection that all hot water piping comply with the insulation requirements in 150.0(J).		
02	Verification Status: ☐ Pass - all applicable requirements are met; or ☐ Fail - one or more applicable requirements are not met. Enter reason for failure in corrections notes field below; or ☐ All N/A - This entire table is not applicable		
03	Correction Notes: <<if Verification Status= Fail, then text entry in this Corrections Notes field is required; user input text>>		
The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met unless otherwise noted in the Verification Status and the Corrections Notes in this table.			

K. ~~HERS~~-Verified Central Parallel Piping Requirements (PP-H) (RA3.6.4)

Systems that utilize this distribution type shall comply with these requirements. <<If A10 or A2-07 = "~~HERS~~-Verified Central Parallel Piping", then display this entire table, else display "section does not apply" message>>

01	Each central manifold has 5 feet or less of pipe between manifold and water heater. (RA4.4.15)	
02	For manifolds that include valves, the manifold must be readily accessible in accordance with the plumbing code. (RA4.4.4)	
03	Hot water distribution system piping from the manifold to the fixtures and appliances must take the most direct path. For example, piping from a second story manifold cannot supply the first floor. (RA4.4.4)	
04	The hot water distribution piping must be separated by at least 2 inches from any other hot water supply piping, and at least 6 inches from any cold water supply piping. Alternatively, the hot water supply piping must be insulated to the thicknesses shown in TABLE 120.3-A-1. (RA 4.4.4)	
05	Verification Status:	<<user pick from list: ☐ Pass - all applicable requirements are met; or ☐ Fail - one or more applicable requirements are not met. Enter reason for failure in corrections notes field below; or ☐ All N/A - This entire table is not applicable
06	Correction Notes: <<if Verification Status= Fail, then text entry in this Corrections Notes field is required; user input text>>	

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met unless otherwise noted in the Verification Status and the Corrections Notes in this table.

L. Central Parallel Piping Requirements (~~PP~~) (RA4.4.4)

Systems that utilize this distribution type shall comply with these requirements. <<If A10 or A2-07 = "Central Parallel Piping", then display this entire table, else display "section does not apply" message>>

01	Each central manifold has 15 feet or less of pipe between manifold and water heater	
02	For manifolds that include valves, the manifold must be readily accessible in accordance with the plumbing code.	
03	Hot water distribution system piping from the manifold to the fixtures and appliances must take the most direct path. For instance, piping from a second story manifold cannot supply the first floor	
04	The hot water distribution piping must be separated by at least 2 inches from any other hot water supply piping, and at least 6 inches from any cold water supply piping. Alternatively, the hot water supply piping must be insulated to the thicknesses shown in TABLE 120.3-A-1.	

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met.

M. Point of Use Requirements (POU) (RA4.4.5)

Systems that utilize this distribution type shall comply with these requirements <<If A10 or A2-07 = "Point of Use", then display this entire table, else display "section does not apply" message>>

01	All hot water supply pipe run lengths are equal to or less than the maximum values shown below, based on the pipe diameter. If a combination of piping is used in a single run, then one half the allowed length of each size is the maximum installed length. The maximum allowed length of piping for the longest run terminating in: 3/8 inch - For only one pipe size - max length allowed is 15 feet For combination pipe sizes the max allowed length of 3/8-inch piping is 7.5 feet, of 1/2 inch piping is 5 feet, and 3/4 inch piping is 2.5 feet. 1/2 inch - For only one pipe size – max length allowed is 10 feet For combination pipe sizes the allowed length of 1/2 inch piping is 5 feet, and 3/4 inch piping is 2.5 feet. 3/4 inch - For only one pipe size = 5 feet	
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The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met.

N. Mandatory Requirements for all Recirculation System (RA4.4.7)

Systems that utilize this distribution type shall comply with these requirements. <<If A10 or A2-07 = "Recirculation System Non-Demand Control", "Demand Recirculation Manual Control", "Demand Recirculation Sensor Control", "~~HERS~~-Verified Demand Recirculation Manual Control", or "~~HERS~~-Verified Demand Recirculation Sensor Control", then display this entire table, else display "section does not apply" message>>

01	A check valve located between the recirculation pump and the water heater to prevent unintentional recirculation.
02	Piping must take most direct path between water heater and fixtures.
03	Insulation is not required on the cold water line when it is used as the return.
04	If more than one loop installed each loop shall have its own pump and controls.
The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met	

O. Recirculation Non-Demand Controls Requirements (~~R-ND~~) (RA4.4.8)

Systems that utilize this distribution type shall comply with these requirements. <<If A10 or A2-07 = "Recirculation System Non-Demand Control, then display this entire table, else display "section does not apply" message >>

01	The active control shall be either: timer, temperature, or time and temperature. Timers shall be set to less than 24 hours. The temperature sensor shall be connected to the piping and to the controls for the pump.
The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met.	

P. Demand Recirculation Manual Control (~~R-DRmc~~) (RA4.4.9)/Sensor Control Requirements (~~RDRsc~~) (RA4.4.10)

Systems that utilize this distribution type shall comply with these requirements. <<If A10 or A2-07 = "Demand Recirculation Manual Control", "Demand Recirculation Sensor Control", "~~HERS~~-Verified Demand Recirculation Manual Control", or "~~HERS~~-Verified Demand Recirculation Sensor Control", then display this entire table, else display "section does not apply" message >>

01	The system operates "on-demand", meaning that the pump begins to operate shortly before or immediately after hot water draw begins, and stops when the return water temperature reaches a certain threshold value. For Demand Recirculation Manual Control, the pump shall be turned on using a manual switch system. For Demand Recirculation Sensor Control, the pump shall be turned on using a sensor system.
02	The controls shall be located in the kitchen, bathroom, and any hot water fixture location that is at least 20 feet from the water heater.
03	Manual controls may be activated by wired or wireless mechanisms. <u>Each control shall have standby power of 1 Watt or less.</u>
04	Sensor Controls may be activated by wired or wireless mechanisms, including buttons, motion sensors, door switches and flow switches. Each control shall have standby power of 1 Watt or less.
05	Pump and control placement shall meet one of the following criteria: - When a dedicated return line has been installed the pump, controls and thermo-sensor are installed at the end of the supply portion of the recirculation loop; or - The pump and controls are installed on the dedicated return line near the water heater and the thermo-sensor is installed in an accessible location as close to the end of the supply portion of the recirculation loop as possible; or - When the cold water line is used as the return, the pump, demand controls and thermo-sensor shall be installed in an accessible location at the end of supply portion of the hot water distribution line (typically under a sink).
06	After the pump has been activated, the controls shall allow the pump to operate until the water temperature at the thermo-sensor rises to one of the following values: - Not more than 10°F (5.6°C) above the initial temperature of the water in the pipe - Not more than 102°F (38.9°C).
07	Controls shall limit operation to no more than 5 minutes following activation.

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met.

Q. ~~HERS~~-Verified Demand Recirculation Manual Control (RDRmc-H) (RA3.6.6)/Sensor Control (RDRsc-H) (RA3.6.7) Requirements

Systems that utilize this distribution type shall comply with these requirements. <<If A10 or A2-07 = "~~HERS~~-Verified Demand Recirculation Manual Control" or "~~HERS~~-Verified Demand Recirculation Sensor Control", then display this entire table, else display "section does not apply" message >>

01	HERS-ECC rater shall perform a visual inspection to verify that the demand pump, manual/sensor controls and thermo-sensor are present and operating properly consistent with the applicable requirements of RA4.4.9 and RA4.4.10	
02	Verification Status:	<<user pick from list: ☐ Pass - all applicable requirements are met; or ☐ Fail - one or more applicable requirements are not met. Enter reason for failure in corrections notes field below; or ☐ All N/A - This entire table is not applicable
03	Correction Notes: <<if Verification Status= Fail, then text entry in this Corrections Notes field is required; user input text>>	

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met unless otherwise noted in the Verification Status and the Corrections Notes in this table.

R. Determination of ~~HERS-ECC~~ Verification Compliance

<< if D08 result = System Complies, and results for all applicable sections F, G, I, J, K, and Q ≠ fail, then display: Complies: All specified verification protocol requirements on this document are met; else display: Does not comply: One or more specified verification protocol requirements on this document are not met.>>